12 A function is defined by $f(x) = x^3 - x$.

(a) By considering $\dfrac{f(x+h) - f(x)}{h}$, show from first principles that $f'(x) = 3x^2 - 1$. **[4]**

(b) Sketch the gradient function $f'(x)$. **[2]**

(c) Show that the curve $y = f(x)$ has a single point of inflection which is not a stationary point. **[3]**

12(a)

12(b)

12(c)